TECHNICAL DOCUMENTATION

EFT

Equity Forecast Terminal

Mathematical & Statistical Foundations

Version 5.1

Core model	Geometric Brownian Motion (GBM)
Calibration	Bayesian Fusion (historical + theoretical priors)
Simulation	Monte Carlo with AR(1) and dynamic volatility
Confidence	Volume · $\hat{\sigma}$ precision · Kurtosis · EWMA · $\sigma\sqrt{T}$
Risk	VaR 95% · CVaR · MC loss prob. · Directional consensus

This document describes the mathematical foundations of each component of the terminal.

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Contents

1 Model Overview

The EFT terminal is built around a four-step processing pipeline:

1. **Calibration:** estimation of parameters (μ, σ) via Bayesian fusion of an empirical source (historical log-returns) and a theoretical source (user-defined scenario probabilities).
2. **Simulation:** generation of price trajectories using an extended GBM with AR(1) autocorrelation and simplified GARCH-type dynamic volatility.
3. **Evaluation:** two independent scores measure the reliability of the model (*Confidence Index*) and the level of investment risk (*Risk Score*).
4. **Presentation:** percentile bands P20/P50/P80, weighted consensus curve, final price distribution, and technical indicators.

2 Parameter Calibration — Bayesian Fusion

2.1 Empirical Log-Returns

Let $\{P_0, P_1, \ldots, P_N\}$ be the historical price series. The daily log-returns are defined as:

$$r_i = \ln\left(\frac{P_i}{P_{i-1}}\right), \quad i = 1, \ldots, N \tag{1}$$

The empirical estimators are:

$$\hat{\mu}_{\text{emp}} = \frac{1}{N}\sum_{i=1}^{N} r_i \tag{2}$$

$$\hat{\sigma}_{\text{emp}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(r_i - \hat{\mu}_{\text{emp}})^2} \tag{3}$$

2.2 Theoretical Prior (User Probabilities)

The user specifies three discrete scenarios with variations v_j and probabilities p_j ($\sum_j p_j = 1$):

$$\mathbb{E}[X] = \sum_{j=1}^{3} v_j\, p_j \tag{4}$$

$$\text{Var}[X] = \sum_{j=1}^{3} v_j^2\, p_j - \big(\mathbb{E}[X]\big)^2 \tag{5}$$

Normalised by the current price P_0:

$$\mu_{\text{theo}} = \frac{\mathbb{E}[X]}{P_0 \cdot \max\!\big(1,\, |\mathbb{E}[X]|\big)}, \qquad \sigma_{\text{theo}} = \frac{\sqrt{\text{Var}[X]}}{P_0} \tag{6}$$

2.3 Bayesian Fusion

Both sources are weighted via a strength parameter $k > 0$:

$$w_{\text{hist}} = \frac{N}{N+k}, \qquad w_{\text{theo}} = 1 - w_{\text{hist}} \tag{7}$$

The fused parameters are:

$$\mu = w_{\text{hist}} \cdot \hat{\mu}_{\text{emp}} + w_{\text{theo}} \cdot \mu_{\text{theo}} \tag{8}$$

$$\sigma = \max\bigl(w_{\text{hist}} \cdot \hat{\sigma}_{\text{emp}} + w_{\text{theo}} \cdot \sigma_{\text{theo}},\ 0.8\,\sigma_{\text{theo}}\bigr) \tag{9}$$

The uncertainty on μ is estimated as $\sigma_\mu = \sigma/\sqrt{N}$.

2.4 Technical Indicators

OLS regression on prices (trend):

$$\hat{\beta}_1 = \frac{\displaystyle\sum_{i=0}^{N}(i-\bar{i})(P_i-\bar{P})}{\displaystyle\sum_{i=0}^{N}(i-\bar{i})^2} \tag{10}$$

Adapted RSI:

$$\text{RSI} = 100 - \frac{100}{1+\bar{G}/\bar{L}} \tag{11}$$

3 Monte Carlo Simulation — Extended GBM

3.1 Geometric Brownian Motion

$$P_{t+1} = P_t \cdot \exp\Bigl[\Bigl(\mu - \tfrac{1}{2}\sigma_t^2\Bigr) + \sigma_t\, z_t\Bigr] \tag{12}$$

The Itô correction $-\frac{1}{2}\sigma_t^2$ ensures $\mathbb{E}[P_{t+1}] = P_t \cdot e^{\mu}$.

3.2 Normal Variates — Box-Muller

$$z = \sqrt{-2\ln(u_1)} \cdot \cos(2\pi u_2) \sim \mathcal{N}(0,1) \tag{13}$$

3.3 AR(1) Extension

$$\tilde{z}_t = z_t + \rho\,\tilde{z}_{t-1}, \qquad |\rho| < 0.95 \tag{14}$$

3.4 Dynamic Volatility — Simplified GARCH

$$\sigma_t = \sigma \cdot \bigl(1 + 0.15 \cdot |\tilde{z}_{t-1}|\bigr) \tag{15}$$

3.5 Scenarios and Confidence Bands

Scenario	Drift	Volatility
Bullish	$\mu + \sigma_\mu$	$0.85\,\sigma$
Neutral (Base)	μ	σ
Bearish	$\mu - \sigma_\mu$	$1.15\,\sigma$

3.6 Consensus Curve

Weighted mixture $(w_B, w_N, w_b) = (0.25,\ 0.50,\ 0.25)$:

$$m_T^{\text{cons}} = \sum_s w_s \cdot m_T^{(s)} \tag{16}$$

$$\sigma_T^{\text{pool}} = \sqrt{\sum_s w_s \cdot \left(\sigma_T^{(s)}\right)^2} \tag{17}$$

4 Confidence Index

The Confidence Index answers the question: *is the model well-calibrated?* It measures the statistical reliability of the estimated parameters, independently of the asset's direction or performance.

$$C = \text{clip}\left(\sum_{i=1}^{5} s_i,\ 1,\ 100\right) \tag{18}$$

#	Component	Weight	Formula
1	Data volume	25%	$1 - e^{-N/60}$
2	$\hat{\sigma}$ precision	25%	$1 - \text{CI}_{95}(\sigma)/\sigma$
3	Fat tails (kurtosis)	20%	$e^{-\sqrt{\lvert\kappa\rvert}\times 0.4}$
4	Stationarity (EWMA)	20%	$1 - \lvert\rho_v - 1\rvert \times 1.2$
5	Temporal decay $\sigma\sqrt{T}$	10%	$e^{-\sigma\sqrt{T}\times 4}$

Table 1: Confidence Index components

4.1 Component 1 — Data Volume (25%)

By the law of large numbers, estimator quality increases with N. This convergence is modelled by a negative exponential:

$$s_1 = 25 \cdot \min\!\left(1,\ 1 - e^{-N/60}\right) \tag{19}$$

N	10	30	60	120	252
Score	13	39	63	86	98

4.2 Component 2 — $\hat{\sigma}$ Precision (25%)

Volatility σ is the key parameter of the GBM. Its 95% confidence interval is:

$$\mathrm{CI}_{95}(\sigma) = \hat{\sigma}_{\mathrm{emp}} \cdot \sqrt{\frac{2}{N-1}} \cdot 1.96 \tag{20}$$

The score measures the relative precision of this estimator:

$$s_2 = 25 \cdot \max\left(0,\ 1 - \frac{\mathrm{CI}_{95}(\sigma)}{\hat{\sigma}_{\mathrm{emp}}}\right) \tag{21}$$

N	10	30	60	120	252
Score	8	49	64	75	83

This component is independent of the asset's direction: it only quantifies whether σ has been observed enough times to be reliable.

4.3 Component 3 — Fat Tails, Excess Kurtosis (20%)

GBM assumes normally distributed residuals. Excess kurtosis κ measures the deviation from this assumption:

$$\kappa = \frac{\frac{1}{N}\sum_i (r_i - \bar{r})^4}{\sigma_{lr}^4} - 3 \tag{22}$$

An exponential decay in $\sqrt{|\kappa|}$ is used — more robust than a simple exponential, it does not vanish even for extreme kurtosis values (typical real values: $\kappa \in [2, 15]$):

$$s_3 = 20 \cdot \exp\left(-\sqrt{|\kappa|} \times 0.4\right) \tag{23}$$

$\lvert\kappa\rvert$	0	1	3	5	10	15
Score	100	67	50	41	28	21

4.4 Component 4 — EWMA Stationarity (20%)

The JP Morgan RiskMetrics model ($\lambda = 0.94$) estimates recent volatility:

$$\hat{\sigma}^2_{\mathrm{EWMA},t} = \lambda\, \hat{\sigma}^2_{\mathrm{EWMA},t-1} + (1-\lambda)\, r_t^2 \tag{24}$$

The ratio $\rho_v = \hat{\sigma}_{\mathrm{EWMA}}/\sigma$ measures regime stability:

$$s_4 = 20 \cdot \max\left(0,\ 1 - |\rho_v - 1| \times 1.2\right) \tag{25}$$

$\rho_v \approx 1$ indicates a stable regime; $\rho_v \gg 1$ signals a recent shock not representative of the historical data.

4.5 Component 5 — Temporal Decay $\sigma\sqrt{T}$ (10%)

Even with perfectly estimated parameters, forecast quality degrades with horizon T and volatility σ:

$$s_5 = 10 \cdot \exp\Big(-\sigma\sqrt{T} \times 4\Big) \tag{26}$$

	$T = 5\,\text{d}$	$T = 20\,\text{d}$	$T = 60\,\text{d}$
$\sigma = 1.5\%$	94	80	68
$\sigma = 3.0\%$	87	64	46

5 Investment Risk Score

The Risk Score answers the question: *what is the probability of losing money on this investment?* It is independent of the Confidence Index — a model can be well-calibrated and still predict a high-risk asset.

$$R = \text{clip}\left(\left[\sum_{i=1}^{4} \frac{1}{4}\, r_i\right] \times m,\ 1,\ 99\right) \tag{27}$$

where m is a bearish convergence multiplier defined below.

#	**Component**	**Weight**	**Nature**
1	Analytical VaR 95%	25%	Analytical (GBM)
2	CVaR / Expected Shortfall	25%	Monte Carlo
3	Loss probability	25%	Monte Carlo
4	Directional consensus	25%	Multi-indicator

Table 2: Risk Score components (equal weights)

5.1 Component 1 — Analytical VaR 95% (25%)

Under the GBM assumption, the maximum probable loss at 95% over horizon T is computed analytically:

$$\text{VaR}_{95}(T) = P_0 \cdot \Big(1 - \exp\Big[\Big(\mu - \tfrac{1}{2}\sigma^2\Big)T - 1.645\,\sigma\sqrt{T}\Big]\Big) \tag{28}$$

The factor $1.645 = \Phi^{-1}(0.05)$ is the 5% quantile of the standard normal distribution. Normalised:

$$r_1 = \min\left(100,\ \frac{\text{VaR}_{95}(T)/P_0}{40\%} \times 100\right) \tag{29}$$

5.2 Component 2 — CVaR / Expected Shortfall 95% (25%)

The CVaR (Conditional Value at Risk), also called Expected Shortfall, is the reference risk measure mandated by Basel III since 2016. It measures the average loss in the worst 5% of cases:

$$\text{CVaR}_{95}(T) = \frac{1}{|Q_{5\%}|} \sum_{i \in Q_{5\%}} \max\Big(0,\ P_0 - P_T^{(i)}\Big) \tag{30}$$

where $Q_{5\%}$ is the set of the 5% worst simulations. Normalised:

$$r_2 = \min\left(100,\ \frac{\mathrm{CVaR}_{95}(T)/P_0}{50\%} \times 100\right) \tag{31}$$

CVaR is more conservative than VaR: where VaR says the loss will not exceed X in 95% of cases, CVaR says that if we are in the worst 5% of cases, the average loss will be Y.

5.3 Component 3 — Monte Carlo Loss Probability (25%)

The fraction of simulations ending below the entry price P_0:

$$\hat{p}_{\mathrm{loss}} = \frac{\#\{P_T^{(i)} < P_0\}}{N_{\mathrm{sim}}} \tag{32}$$

$$r_3 = \hat{p}_{\mathrm{loss}} \times 100 \quad \text{(direct scale } 0\% \to 100\%\text{)} \tag{33}$$

5.4 Component 4 — Directional Consensus (25%)

Five independent indicators are consulted. Each votes bearish or not:

Indicator	**Bearish condition**
Fused drift μ	$\mu < 0$
OLS trend $\hat{\beta}_1$	$\hat{\beta}_1 < 0$
RSI	RSI < 50
MC loss probability	$\hat{p}_{\mathrm{loss}} > 50\%$
Annualised drift	$\mu \times 252 < -5\%$

b = number of bearish votes $\in \{0, \dots, 5\}$, $\quad r_4 = \dfrac{b}{5} \times 100$

5.5 Bearish Convergence Multiplier

When the conditions $b \geq 4$, $\hat{p}_{\mathrm{loss}} > 65\%$, and $\mu < 0$ are simultaneously satisfied, a multiplier amplifies the score upward:

$$m = \begin{cases} 1 + \dfrac{b}{5} \times 0.20 & \text{if } b \geq 4,\ \hat{p}_{\mathrm{loss}} > 65\%,\ \mu < 0 \\ 1 & \text{otherwise} \end{cases} \tag{34}$$

5.6 Verdicts and Thresholds

Score	**Verdict**	**Interpretation**
≥ 70	Very High	Loss likely regardless of outcome
≥ 50	High	Significant risk, multiple bearish signals
≥ 30	Moderate	High uncertainty, no clear direction
≥ 15	Low	Mostly bullish profile, loss unlikely
< 15	Very Low	Clear upward trend

6 Final Price Distribution

Under the GBM assumption, the price P_T follows a log-normal distribution:

$$\ln(P_T) \sim \mathcal{N}\Big(\ln(P_0) + \Big(\mu - \tfrac{1}{2}\sigma^2\Big)T,\ \sigma^2 T\Big) \tag{35}$$

EFT estimates the distribution empirically over $N_{\text{dist}} = \min(N_{\text{sim}}, 1000)$ trajectories. The loss probability:

$$\hat{p}(P_T < P_0) = \frac{1}{N_{\text{dist}}} \sum_{i=1}^{N_{\text{dist}}} \mathbf{1}_{\{P_T^{(i)} < P_0\}} \tag{36}$$

7 Bollinger Bands

With a rolling window of $m = 20$ days:

$$\text{MA}_t = \frac{1}{m} \sum_{i=t-m+1}^{t} P_i \tag{37}$$

$$\text{BB}_{\pm}(t) = \text{MA}_t \pm 2\,\hat{\sigma}_t^{\text{Boll}} \tag{38}$$

8 Summary — Properties and Limitations

8.1 Guaranteed Properties

- **Price positivity:** $P_t > 0$ for all t.
- **Reproducibility:** LCG generator with fixed seed.
- **Bayesian consistency:** $w_{\text{hist}} + w_{\text{theo}} = 1$.
- **Itô correction:** $\mathbb{E}[P_{t+1} \,|\, P_t] = P_t \cdot e^{\mu}$.
- **Independence of the two scores:** the Confidence Index measures model reliability; the Risk Score measures investment danger. The two are orthogonal by construction.

8.2 Limitations and Assumptions

- **Stationarity:** μ and σ are assumed constant over the horizon.
- **Fat tails:** excess kurtosis is measured and factored into the Confidence Index, but not integrated into the simulation itself.
- **Liquidity:** bid-ask spreads and transaction costs are ignored.
- **Fundamental data:** none (P/E ratio, earnings, debt).
- **Asset independence:** no cross-asset correlation.

Code variable	**Symbol**	**Description**
`mu`	μ	Fused daily drift
`sigma`	σ	Fused daily volatility
`mu_uncert`	σ_μ	Uncertainty on μ ($= \sigma/\sqrt{N}$)
`w_hist`	w_{hist}	Historical source weight
`rsi`	RSI	Relative Strength Index
`slope`	$\hat{\beta}_1$	OLS slope
`ac`	$\hat{\rho}$	AR(1) autocorrelation
`P0`	P_0	Last historical price
`N`	N	Number of log-returns
`T`	T	Forecast horizon (days)

Table 3: Terminal variables and mathematical notation

8.3 Variable Correspondence Table

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This terminal was designed and developed in cooperation with Claude (Anthropic), an AI assistant that contributed to the mathematical modelling, code development, and writing of this documentation.